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Abstract

Recent advancements in Large Language Models (LLMs) have resulted in the emergence of two distinct model paradigms: **thinking models**, which perform explicit reasoning before producing outputs, and **instruct models**, which generate direct responses based on user instructions. This study presents a comparative evaluation of four representative models, namely o3-mini, GPT-4o mini, DeepSeek-R1, and Llama 3.1 8B. The models were assessed across mathematical reasoning, code generation, general knowledge question answering, and mixed-task scenarios. Performance was evaluated using accuracy and response latency. The experimental findings indicate that while thinking models demonstrate advantages in complex reasoning tasks, instruct models achieve comparable accuracy with lower latency for many practical applications. The results suggest that model selection should be guided by task requirements rather than model category alone.

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1. Introduction

Large Language Models have become increasingly important across domains such as software development, education, customer support, and knowledge management. Modern LLMs can broadly be classified into **thinking models** and **instruct models**. Thinking models allocate additional computational resources to intermediate reasoning steps before generating answers, whereas instruct models prioritize direct instruction-following and efficient response generation.

Although reasoning-oriented architectures are often assumed to provide superior performance, the extent of their advantage across different task categories remains an active area of investigation. Therefore, this study aims to evaluate whether thinking models consistently outperform instruct models and to identify scenarios in which each model type is most effective.

Four models were selected for comparison:

- **Thinking Models:** o3-mini and DeepSeek-R1
- **Instruct Models:** GPT-4o mini and Llama 3.1 8B

The evaluation focused on three primary criteria:

1. Accuracy of generated responses.
2. Response latency.
3. Task suitability across different application domains.

2. Experimentation

2.1 Experimental Setup

The experiments were conducted over three phases using standardized datasets and controlled prompts. All selected models were evaluated using identical tasks to ensure consistency and fairness in comparison.

Phase 1: Preliminary Task Evaluation

A set of ten questions was developed to assess model performance in four categories:

- Factual Question Answering
- Mathematical Reasoning
- Conceptual Understanding
- Code Generation

This initial phase provided baseline observations regarding model correctness and response characteristics.

Phase 2: Mathematical Reasoning

The GSM8K benchmark consisting of fifty grade-school mathematical word problems was used to evaluate reasoning capability. Models were instructed to provide step-by-step solutions followed by a final numerical answer. Accuracy was determined by comparing generated answers with the benchmark solution set.

Phase 3: Code Generation and General Knowledge

Two additional evaluations were performed:

HumanEval Benchmark

- Twenty-five coding problems were selected.
- Generated solutions were tested using the official unit test suite.

General Knowledge Question Answering

- Fifty factual questions were administered.
- Responses were manually evaluated for correctness.

2.2 Evaluation Metrics

The primary evaluation metrics were:

Accuracy

$$Accuracy(\%) = \frac{Correct\ Answers}{Total\ Questions} \times 100$$

Accuracy represented the percentage of correctly solved tasks according to benchmark criteria.

Latency

Latency was measured as the average response time required by a model to generate an answer. This metric reflects computational efficiency and practical usability.

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3. Results and Discussion

3.1 Mathematical Reasoning Performance

The GSM8K benchmark highlighted notable differences in reasoning capability.

Model	Type	Accuracy	Average Latency
o3-mini	Thinking	~98%	2.59 s
GPT-4o mini	Instruct	~96%	2.65 s
Llama 3.1 8B	Instruct	80%	11.51 s
DeepSeek-R1	Thinking	28%	26.32 s

The o3-mini model achieved the highest mathematical reasoning accuracy while maintaining response times comparable to GPT-4o mini. The results suggest that reasoning-oriented architectures may provide measurable benefits for multi-step problem-solving tasks. However, the performance gap between o3-mini and GPT-4o mini remained relatively small, indicating that high-quality instruct models can also effectively handle mathematical reasoning tasks.

3.2 Code Generation Performance

The HumanEval benchmark results are summarized below.

Model	Accuracy	Average Latency
o3-mini	100%	3.50 s
GPT-4o mini	100%	1.98 s
Llama 3.1 8B	100%	110.33 s
DeepSeek-R1	4%	336.00 s

Three models achieved perfect accuracy on the selected HumanEval tasks. Among them, GPT-4o mini demonstrated the lowest latency, producing correct solutions in nearly half the time required by o3-mini. These findings indicate that instruct models can match reasoning models in coding performance while offering superior efficiency.

3.3 General Knowledge Question Answering

The factual question-answering evaluation produced the following results.

Model	Accuracy	Average Latency
Llama 3.1 8B	98%	0.92 s
o3	94%	1.54 s
GPT-4o mini	90%	0.94 s
DeepSeek-R1	50%	16.01 s

Llama 3.1 8B achieved the highest accuracy while maintaining exceptionally low latency. GPT-4o mini also exhibited strong performance, demonstrating that instruct models are highly effective for general-purpose question-answering applications.

3.4 Overall Performance Analysis

Across all evaluated tasks, o3-mini consistently achieved high accuracy scores ranging from 94% to 100%. GPT-4o mini demonstrated comparable performance while maintaining lower response times. Llama 3.1 8B performed particularly well in factual question answering and coding tasks. In contrast, DeepSeek-R1 showed substantially lower performance due to execution constraints and inference limitations within the experimental environment.

The results indicate that task complexity alone does not determine the superiority of thinking models. For many practical applications, instruct models provide similar accuracy with improved efficiency.

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4. Recommendations

Based on the experimental findings, the following recommendations are proposed.

Thinking Models

Thinking models should be preferred when:

- Tasks require multi-step reasoning.
- Mathematical problem solving is a primary requirement.
- Maximum accuracy is prioritized over inference cost.
- Complex analytical decision-making is involved.

Recommended Model: o3-mini, owing to its consistently high accuracy across all evaluated benchmarks.

Instruct Models

Instruct models should be preferred when:

- Fast response times are required.
- Large-scale deployment demands cost efficiency.
- General question answering and content generation are the primary use cases.
- Coding assistance requires both speed and reliability.

Recommended Models:

- GPT-4o mini for general-purpose deployment.
- Llama 3.1 8B for factual question answering and private or offline implementations.

Task-Specific Recommendations

Task Category	Recommended Model

Mathematical Reasoning	o3-mini
Code Generation	GPT-4o mini
General Question Answering	Llama 3.1 8B
Daily Mixed Workloads	GPT-4o mini
Complex Reasoning Tasks	o3-mini
Offline Deployment	Llama 3.1 8B

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5. Conclusion

This study investigated the relative performance of thinking models and instruct models across reasoning, coding, and factual knowledge tasks. The empirical results demonstrate that thinking models are not universally superior to instruct models. While o3-mini achieved the strongest overall reasoning performance, GPT-4o mini and Llama 3.1 8B delivered comparable accuracy with significantly lower latency in several task categories.

The findings further indicate that practical considerations such as response speed, deployment cost, and computational resources play a critical role in model selection. Consequently, organizations and practitioners should evaluate models according to task-specific requirements rather than relying solely on the distinction between thinking and instruct architectures.

In conclusion, the results support the view that both model categories offer distinct advantages. Thinking models are most suitable for complex reasoning-intensive workloads, whereas instruct models provide an effective balance between accuracy, efficiency, and scalability for a wide range of real-world applications.